Assignment Week 5 (Full mark 15)

Q1. Identify the true and false statements. $1 \times 4 = 4$

I. Determination of transport number can be done with the help of Hittorf's and moving boundary methods.

(a) T (b) F

II. The unit of cell constant is cm^2

(a) T (b) F

III. The value of specific counductance (1) is independent of Concentration and temperature.

(a) T (b) F

IV. the transport of Current in salution by the ions depends on charge, concentration and speed of the ions.

(a) T (b) F

Q2. A solution of $AgNO_3$ containing 0.00739 g of $AgNO_3$ per gram of water is electrolyzed between the electrodes. During the experiment 0.078g of Ag was deposited. At the end of the experiment, the anode contains 23.14g of water and 0.236g of $AgNO_3$. Calculate the transport number (t_+) of Ag ions. 2

a) 0.46
b) 0.54
c) 0.38
d) 0.62

Q3. Which one of the following expression regarding transport number is correct.

Where, t_+ is the transport number of cation and t_- is the transport number of anion.

2

a) t_- = Total Current Carried by both the ions /Current Carried by the anion

b) **t_- = Current Carried by the anions / Total Current Carried by both the ions**

c) t_+ = Total Current Carried by both the ions / Current Carried by the cations

d) t_+ = Current Carried by the anions / Total Current Carried by both the ions.

2

Q4. Which of the following statement is correct.

a) The conductance of a solution depends on the number of ions present in the solution.

b) The conductance of a solution is denoted by λ

c) The conductance of a solution is temperature dependents

d) a & c.

Q5. Specific conductance (1) of a solution of KCl is 0.0112 mho cm^{-1}. The resistance of a cell containing the found to be 56. what is the cell constant (cm^{-1}).

2

a) 0.627

b) 0.112

c) 0.427

d) 0.780

Q6. the fraction of the total current carried by an ion is called its **2**

a) Ionic mobility
b) Limiting ionic Conductance
c) **<u>Transport number</u>**
d) Equivalent conductance.

Q7. The specific conductance expressed in **1**

a) **<u>mho cm^{-1}</u>**
b) mho cm
c) mho^{-1} cm^{-1}
d) Ohm^{-1} cm